

S3 Auto Copy Demo

To Begin with the Lab

Summary of the Lab

In this lab, the **Amazon Redshift S3 Auto Copy** feature was configured to automatically load files from **Amazon S3** into a Redshift table whenever new files were uploaded. A preview-mode **Amazon Redshift** provisioned cluster was created because S3 Auto Copy was available only in preview clusters. An S3 bucket and folder were configured as the source location, and a sales table was created as the target table. A COPY job with AUTO mode enabled was then created to monitor S3 events and load matching files automatically. Finally, multiple sales files were uploaded to S3, and system tables were queried to verify that the files were loaded successfully into Redshift without manually running COPY commands.

Understanding Amazon Redshift S3 Auto Copy

Normally, data engineers upload files to **Amazon S3** and manually execute COPY commands to load data into **Amazon Redshift**. This process works well but becomes inefficient when files arrive continuously throughout the day.

Amazon Redshift S3 Auto Copy solves this problem by automatically monitoring an S3 location and loading newly uploaded files into a target Redshift table. Once the auto copy job is created, Redshift listens for S3 events and triggers data loads automatically.

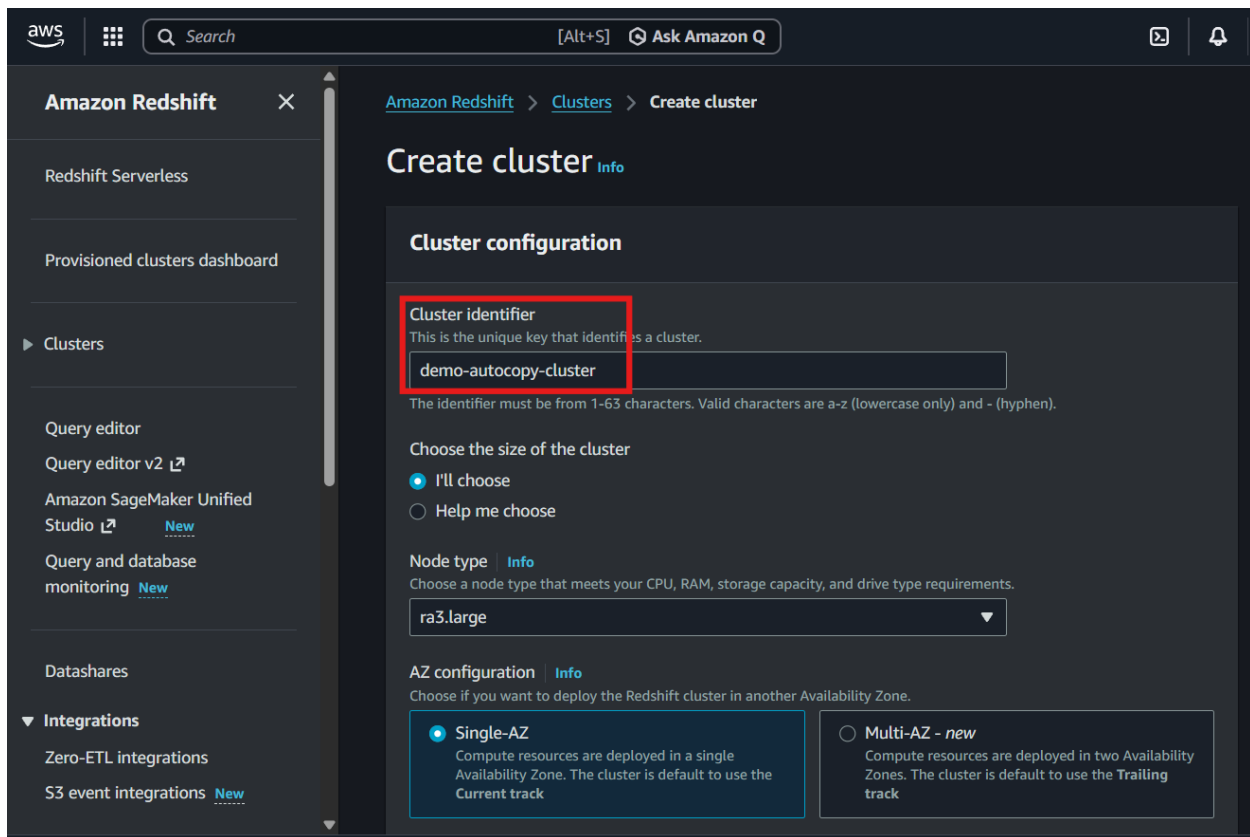
Example:

Consider an e-commerce company receiving sales files every hour from multiple stores. Instead of running COPY commands 24 times a day, an Auto Copy job watches the S3 folder. Whenever a new sales file arrives, Redshift automatically loads the data into the sales table, ensuring dashboards and reports always contain the latest information.

Lab Steps

Create a Redshift Cluster

- Sign in to the AWS Management Console
- Navigate to **Amazon Redshift**
- Open the **Provisioned Clusters Dashboard**
- Click **Create Cluster**
- Enter a cluster name



- Select **ra3.large** as the node type
- Set the number of nodes to **2**
- Disable sample data loading
- Keep the default database username
- Click **Generate password**
- Associate the required IAM roles
 - Default Redshift service role
 - S3 access role (create manually)

Associate IAM roles



IAM roles (1/8)

Choose from existing IAM roles. You can associate up to 50 IAM roles with this cluster.

Q redsh



8 matches

< 1 >



IAM roles



[AmazonRedshift-CommandsAccessRole-20260520T104456](#)



[AmazonRedshift-CommandsAccessRole-20260523T074023](#)



[AmazonRedshift-CommandsAccessRole-20260523T145015](#)



[AmazonSageMakerAdminIAMExecutionRole](#)



[AWSServiceRoleForRedshift](#)



[redshift-spectrum-role](#)



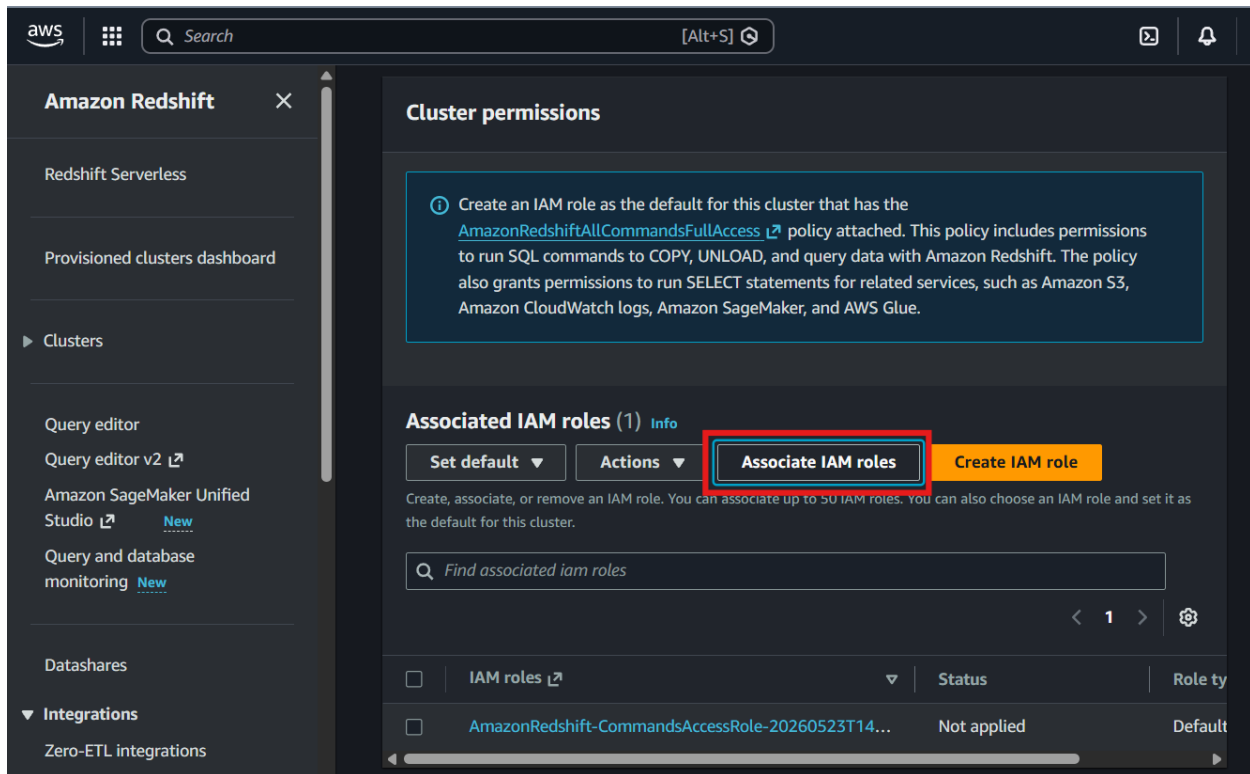
[test-redshift-role-s3](#)



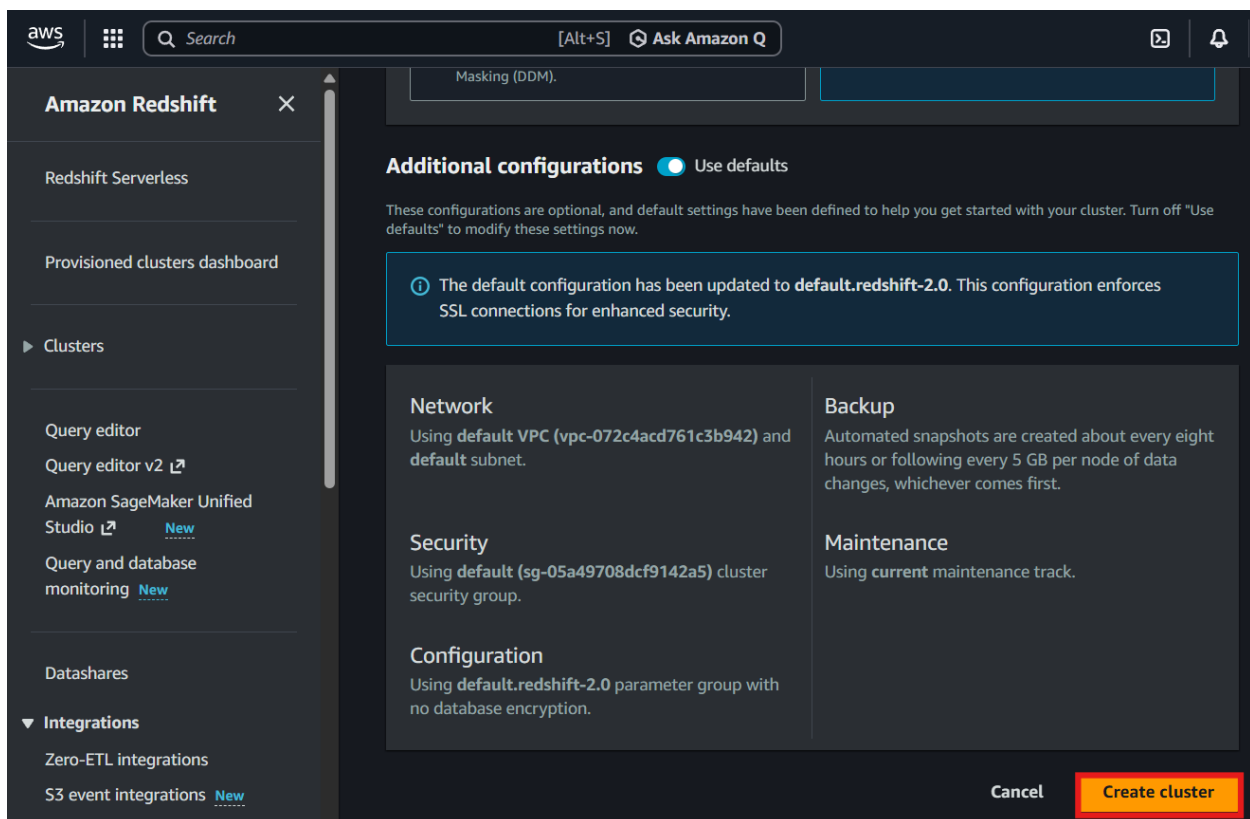
[UserRedShiftroleOregon](#)

Cancel

Associate IAM roles



- Select **Use defaults** for networking configuration
- Click **Create cluster**



- Wait for the cluster status to become **Available**

Create the Source Location in Amazon S3

- Navigate to **Amazon S3**
- Open the bucket used for the lab or create a new one with any name but I am making a new one with **qtd-redshift-demo**

The screenshot shows the 'Create bucket' page in the Amazon S3 console. The left sidebar contains navigation links for Buckets, Files, Access management and security, and Storage management and insights. The main content area is titled 'General configuration'. It shows the AWS Region as 'Asia Pacific (Mumbai) ap-south-1'. Under 'Bucket type', 'General purpose' is selected. Under 'Bucket namespace', 'Global namespace' is selected. The 'Bucket name' field is highlighted with a red box and contains the text 'qtd-redshift-demo'. Below this, there is a 'Choose bucket' button and a note about bucket naming rules. The bottom of the page shows the 'Account and organization' section.

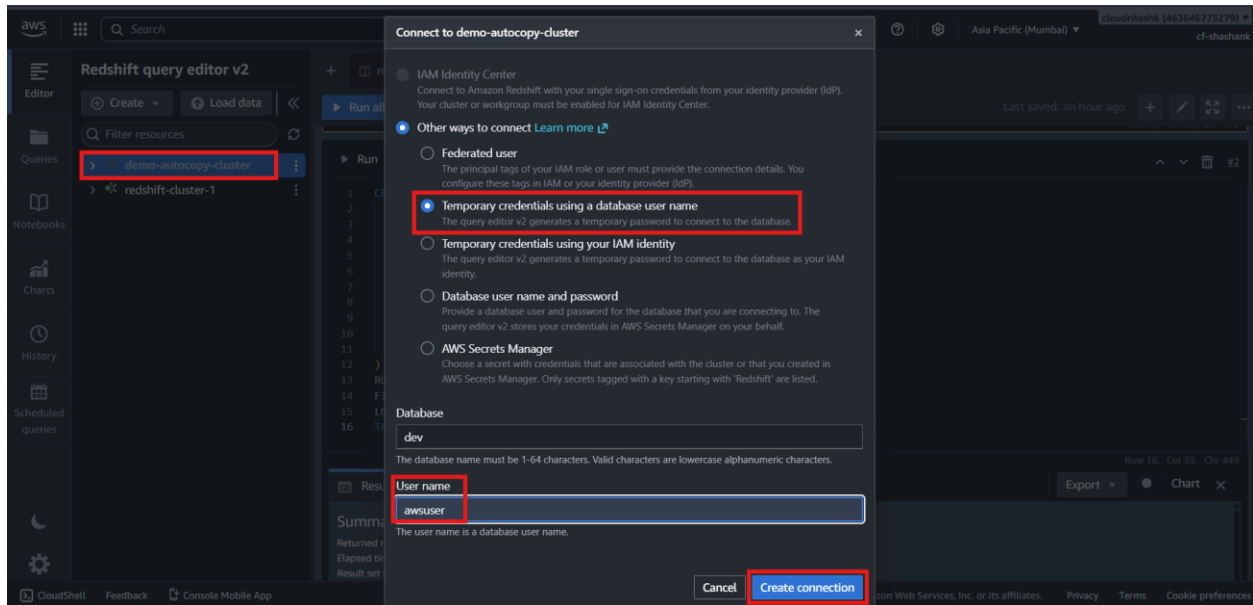
- Click **Create folder**
- Enter the folder name: **auto-copy**

The screenshot shows the 'Create folder' page in the Amazon S3 console. The left sidebar is the same as the previous screenshot. The main content area is titled 'Create folder'. It includes a warning message about bucket policy restrictions. The 'Folder name' field is highlighted with a red box and contains the text 'auto-copy'. Below this, there is a 'Server-side encryption' section with a note about encryption settings. The bottom of the page shows the 'Account and organization' section.

- Click **Create folder**
- Verify that the folder is created successfully

Connect to Query Editor v2

- Navigate to **Amazon Redshift**
- Click **Query data**
- Open **Query Editor v2**
- Create a connection using temporary credentials
- Select the target cluster
- Click **Create connection**

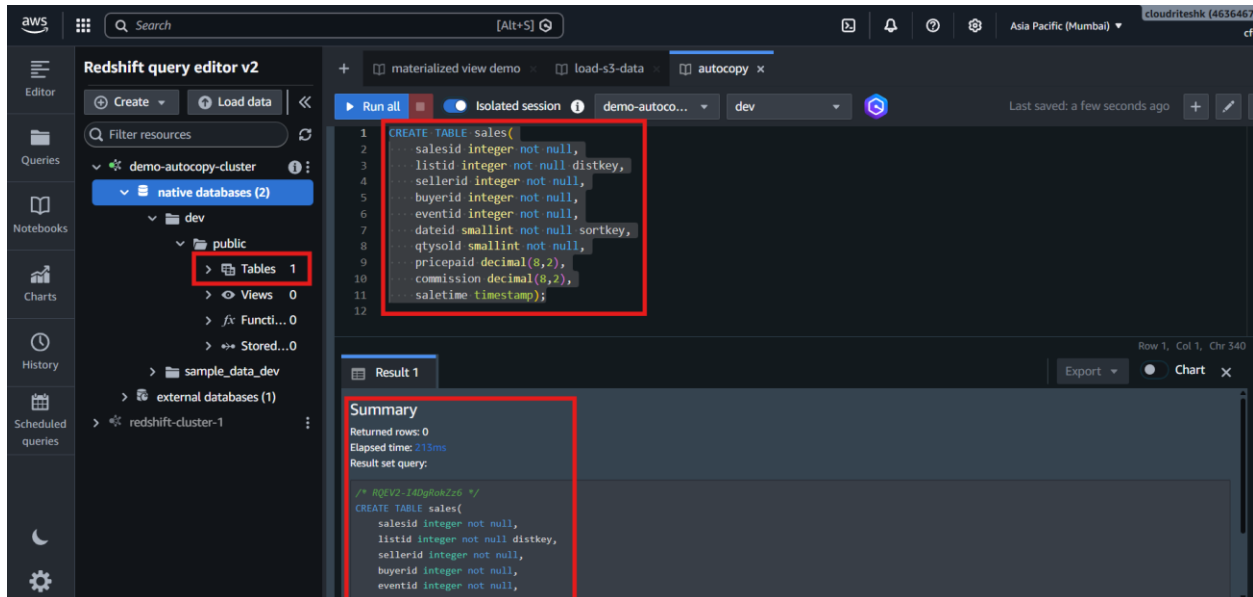


Create the Target Sales Table

- Run the table creation script

```
CREATE TABLE sales(
    salesid integer not null,
    listid integer not null distkey,
    sellerid integer not null,
    buyerid integer not null,
    eventid integer not null,
    dateid smallint not null sortkey,
    qty sold smallint not null,
    price paid decimal(8,2),
    commission decimal(8,2),
    sale time timestamp);
```

- Click **Run**
- Refresh the database explorer
- Verify that the sales table is created successfully



Create the Auto Copy Job

- Create the Auto Copy job using the COPY command with AUTO mode enabled

```
COPY sales
from 's3://{s3_location}/'
iam_role 'arn:aws:iam::${account_id}:role/test-role-redshift-s3'
delimiter '\t'
timeformat 'MM/DD/YYYY HH:MI:SS'
region 'ap-south-1';
JOB CREATE test_auto_copy_sales_job
AUTO ON;
```

- Click **Run**
- Verify that the job is created successfully

Verify Job Creation

- Query the auto copy job metadata

```
SELECT *
FROM sys_copy_job;
```

- Click **Run**
- Verify that the job appears in the system table
- Note the generated Job ID

Verify Initial Table State

- Check the current row count

```
SELECT COUNT(*)
FROM sales;
```

- Verify that the table contains zero records

Upload the First File

- Navigate to **Amazon S3**
- Open the autocopy folder
- Upload **sales1.txt**
- Wait for the upload to complete

Validate the First Auto Load

- Query the auto copy job status

```
SELECT *
FROM sys_copy_job;
```

- Verify that the load status is successful
- Check the sales table row count

```
SELECT COUNT(*)
FROM sales;
```

- Verify that approximately 60,000 records are loaded

Upload the Second File

- Upload **sales2.txt** to the same S3 folder
- Wait for the upload to complete

Validate the Second Auto Load

- Check the sales table row count

```
SELECT COUNT(*)
FROM sales;
```

- Verify that approximately 120,000 records are present
- Query load history and verify two successful load entries

Upload the Third File

- Upload **sales3.txt** to the same S3 folder
- Wait for the upload to complete

Validate the Final Auto Load

- Check the sales table row count

```
SELECT COUNT(*)
FROM sales;
```

- Verify that approximately 172,456 records are loaded
- Query load history

```
SELECT *
FROM stl_load_commits;
```

- Verify that three separate load entries exist
- Verify that each uploaded file generated a separate load event
- Confirm that all three files were loaded automatically without manually executing additional COPY commands

Final Verification

- Verify that the Auto Copy job is active
- Verify that files uploaded to Amazon S3 are loaded automatically into Amazon Redshift

- Verify that system tables record each load operation
- Confirm that Amazon Redshift S3 Auto Copy successfully automates file ingestion from Amazon S3 into Redshift tables.